A Project Report

On

# Atmospheric Water Generator

BY

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**2022A4PS1491H**

Under the supervision of

**Dr. Mrinal Ketan Jagirdar**

**SUBMITTED IN PARTIAL FULLFILLMENT OF THE REQUIREMENTS OF**

**ME F376 (377): DESIGN PROJECT**

**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI (RAJASTHAN)**

**HYDERABAD CAMPUS**

**( 2024-2025 Sem-1)**

# Acknowledgement

I would like to extend special thanks to Dr. Mrinal Ketan Jagirdar Sir for his unwavering support, mentorship, and guidance while carrying out this project. His vast knowledge in the area and valuable pieces of advice have been quite invaluable and really helped me navigate my way through the uncertainties and nuances of designing and implementing the Atmospheric Water Generator system. Sir has always been known for his insightful feedback, and he always encouraged me to cross the boundaries that limit my work. I am grateful for his time and effort devoted to my development as both a student and a researcher.

I would also like to thank Birla Institute of Technology and Science, Pilani-Hyderabad Campus, for providing these resources and an academic environment that has helped the process of this project. The institute has always been very innovative and strong on the research front, and it has contributed in its way towards developing this project to its advancements.

Apart from the above, I would like to thank my peers and colleagues whose feedback and collaboration at different stages of this project were invaluable. Finally, thanks to family and friends who have been continually encouraging and tolerating this hectic schedule and have become a source of strength in the process.

**Birla Institute of Technology and Science-Pilani,**

**Hyderabad Campus**

## Certificate

This is to certify that the project report entitled "**Atmospheric Water Generator**" submitted by Mr. Aditya Varma (ID No. 2022A4PS1491H) in partial fulfillment of the requirements of the course BIO F266, Study Project Course, embodies the work done by him under my supervision and guidance.

**Date:**

**(Dr. Mrinal Ketan Jagirdar)**

BITS- Pilani, Hyderabad Campus

# ABSTRACT

This MATLAB programme has been developed in the framework of a design project for an "Atmospheric Water Generator" (AWG) device, which extracts water from humid ambient air by cooling it below its dew point. In order to optimize the performance of the AWG, the precise calculation and analysis of psychrometric properties are essential. Such a program will enable one to calculate a complete set of psychrometric properties conditioned by three fundamental inputs: height above sea level, in order to make possible the definition of total atmospheric pressure, and two others like temperature, humidity ratio, or relative humidity.

In addition to its ability to return numerical results, the program offers powerful graphical capabilities. It can plot constant property curves on a psychrometric chart that will be used to understand how the various air parameters interact at different conditions. The program also allows plotting of 3D curves based on one psychrometric property to intuitively recognize any trends or correlations. Features such as these are most beneficial in emulating varied environmental conditions to optimize the design and operational efficiency of the AWG.

The system can also be used as a versatile input configuration tool: flexible and easy to use and therefore very useful for researchers and engineers in the field of air conditioning, dehumidification, and water extraction technology. Through computing and visualizing accurately the properties of atmospheric air, this MATLAB tool is part of the development and fine-tuning process of the AWG device.

# CONTENTS

# Chapter-1

## Implementation of ASHRAE tables

While going at temperatures below 0 Degree Celsius, the formula for finding Saturation Temperature when Pressure is given was giving errors as it is specified in the “Heating And Cooling Of Buildings” textbook that it is only valid for temperatures between 0 Degree Celsius and 93 Degree Celsius. Also, to increase the range of temperatures and pressures, I have implemented ASHRAE tables for finding Saturation Pressure when Temperature is given and Saturation Temperature when Pressure is given while eliminating the use of the conversion formulae.

```
function P_sat = getSaturationPressure(T)

   T_data = [Data from ASHRAE tables];

   P_data = [Data from ASHRAE tables];

   P_sat = interp1(T_data, P_data, T, 'pchip', 'extrap');

end

function T_sat = getSaturationTemperature(P)

   T_data = [Data from ASHRAE tables];

   P_data = [Data from ASHRAE tables];

   T_sat = interp1(P_data, T_data, P, 'pchip', 'extrap');

end
```

We are using PCHIP method of interpolation as it is the most suitable method for interpolation of steam tables.

# Chapter-2

## Solved all the problems occurring during Iterative Analysis

I have checked for different end points and locations where the iterative analysis were not converging to a point and tried to fix the functions.

Also, I have implemented the iterative functions using ASHRAE tables which makes them concise.

# Chapter-3

## Implemented a GUI version of the program

To make the program more user friendly, I have implemented a GUI version of the program.

To make this possible, I have converted the MATLAB code into Python code as Python comes with built-in libraries that make it easier to implement GUI. Also, Python also comes with libraries such as Matplotlib and NumPy that are like MATLAB and helps in plotting graphs and doing complex calculations.

I have also added some features to the program that will be helpful in real world applications and help us in managing the datapoints from our AWG and other projects that require the calculations of Psychrometric properties.

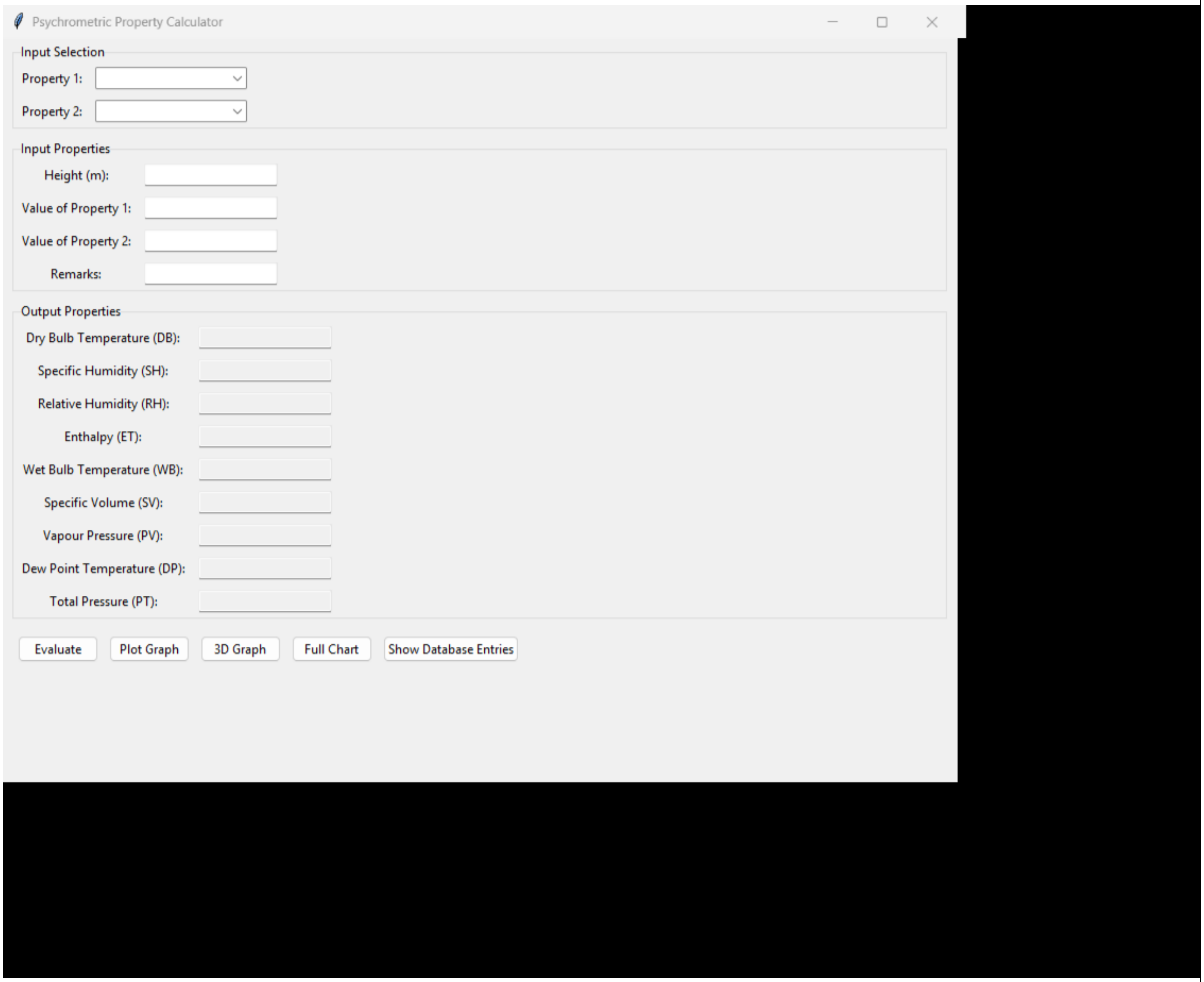
Here are the features I added-

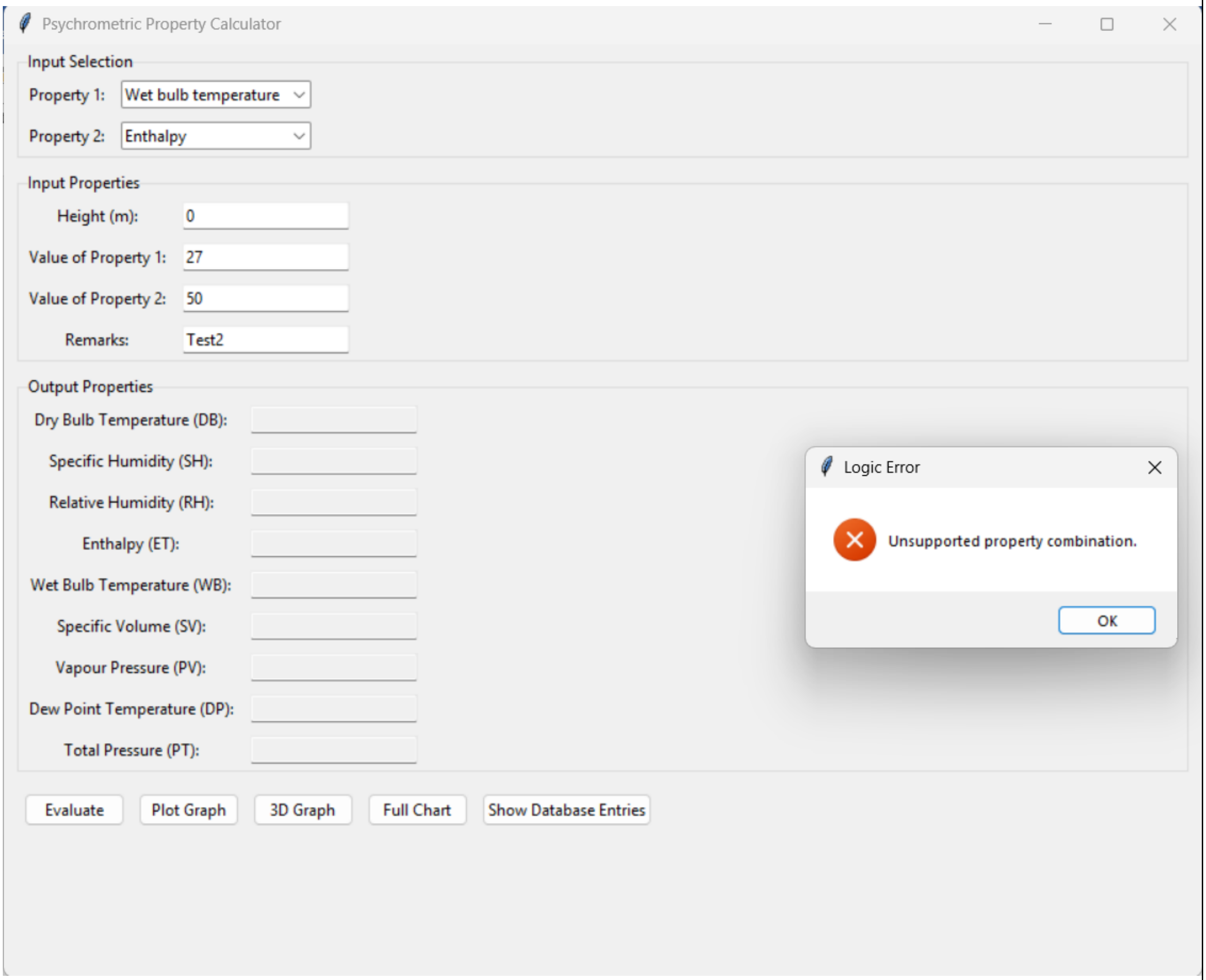
1. A Graphical interface that includes a dialog box that guides the user to select any two Psychrometric properties and then enter their values along with height from sea level and remarks for that calculation. After entering the properties, when the user clicks on

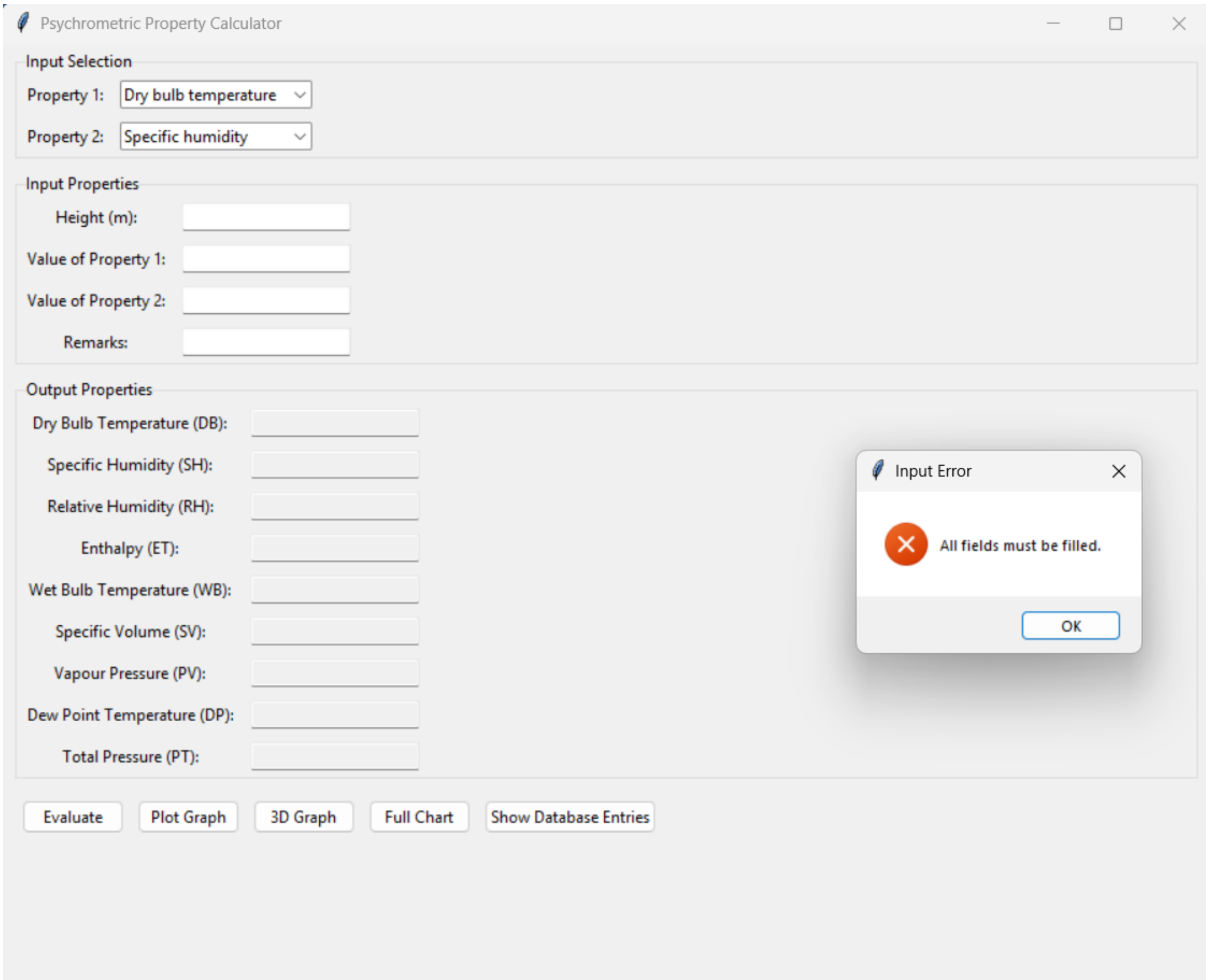
Evaluate, the program calculates all the Psychrometric properties and displays them on the screen. The GUI also helps the user in selecting the Psychrometric properties so that the user does not select any incompatible set of Psychrometric properties. Also, the GUI does not allow the user to click on Evaluate without entering the values of both the Psychrometric properties and height from seal level.

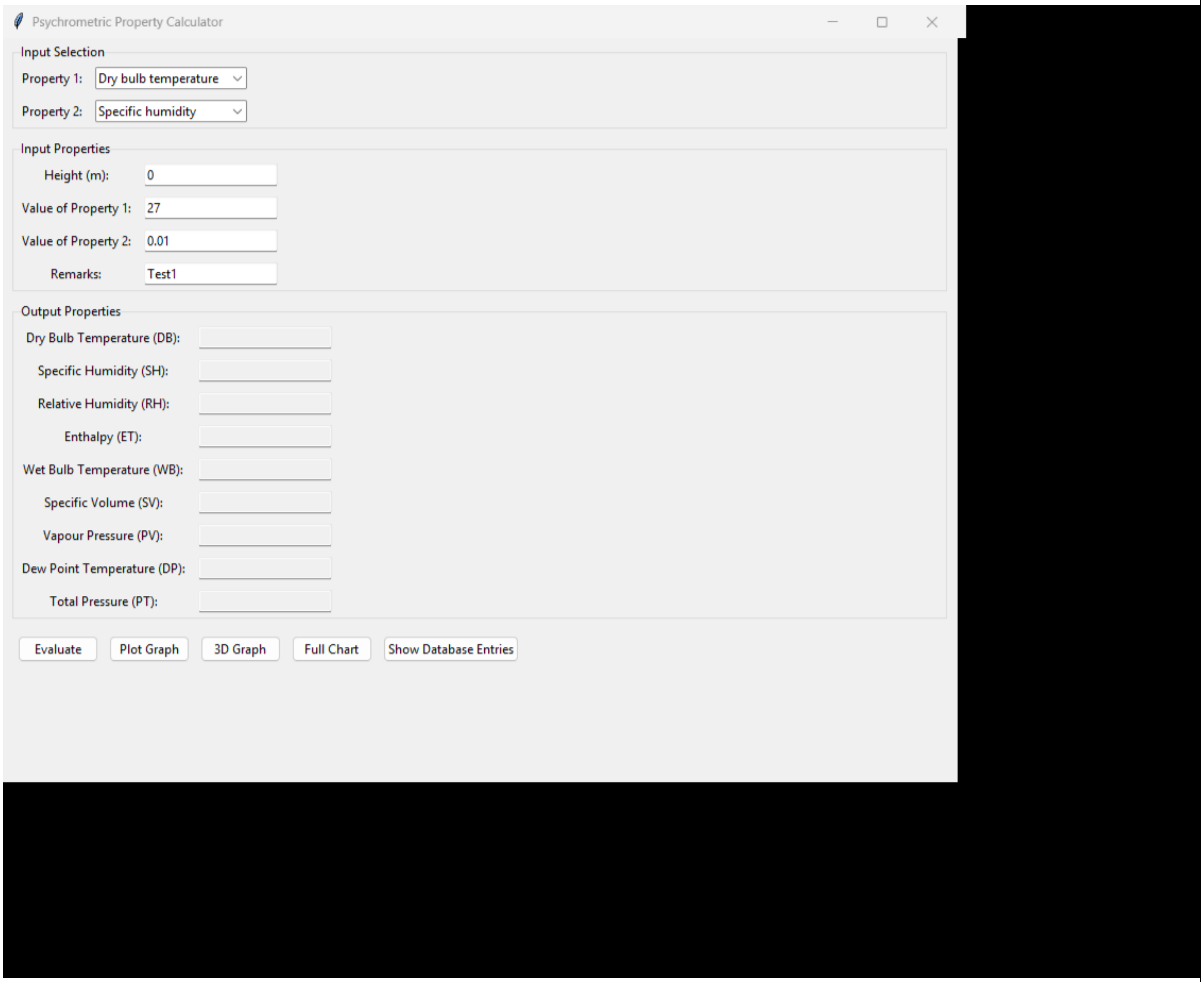
2. Once the user clicks on Evaluate and all the Psychrometric properties have been calculated, the program automatically enters all the data into a database along with the remarks and System time. Which helps managing the data of our calculations.
3. Once the user clicks on Evaluate, the Plot Graph button is enabled. Once the user clicks on it, a new dialog-box opens and plots constant property lines of all the calculated Psychrometric properties which also gives an option to save the graph as in image. Also, the GUI does not allow the user to click on Plot Graph button before the user clicks on Evaluate button.
4. The GUI also provides a button 3D Graph. Once the user clicks on 3D Graph, a new dialog-box opens and asks the user for selecting any Psychrometric property and its value. Once the values is entered, a new dialog-box opens and plots a constant property curve of that Psychrometric property which also gives an option to save the plot as an image.
   **(This feature is under development)**
5. The GUI also provides a button Full Graph. Once the user clicks on Full Graph, a new dialog-box opens and asks the user for height from sea level. Once the value is entered, a new dialog-box opens and plots the full Psychrometric chart which also gives an option to save the graph as an image.
6. The GUI also provides a button Show Database Entries. Once the user clicks on Show Database Entries, a new dialog-box opens and fetches all the previous calculations in different rows along with remarks and date.

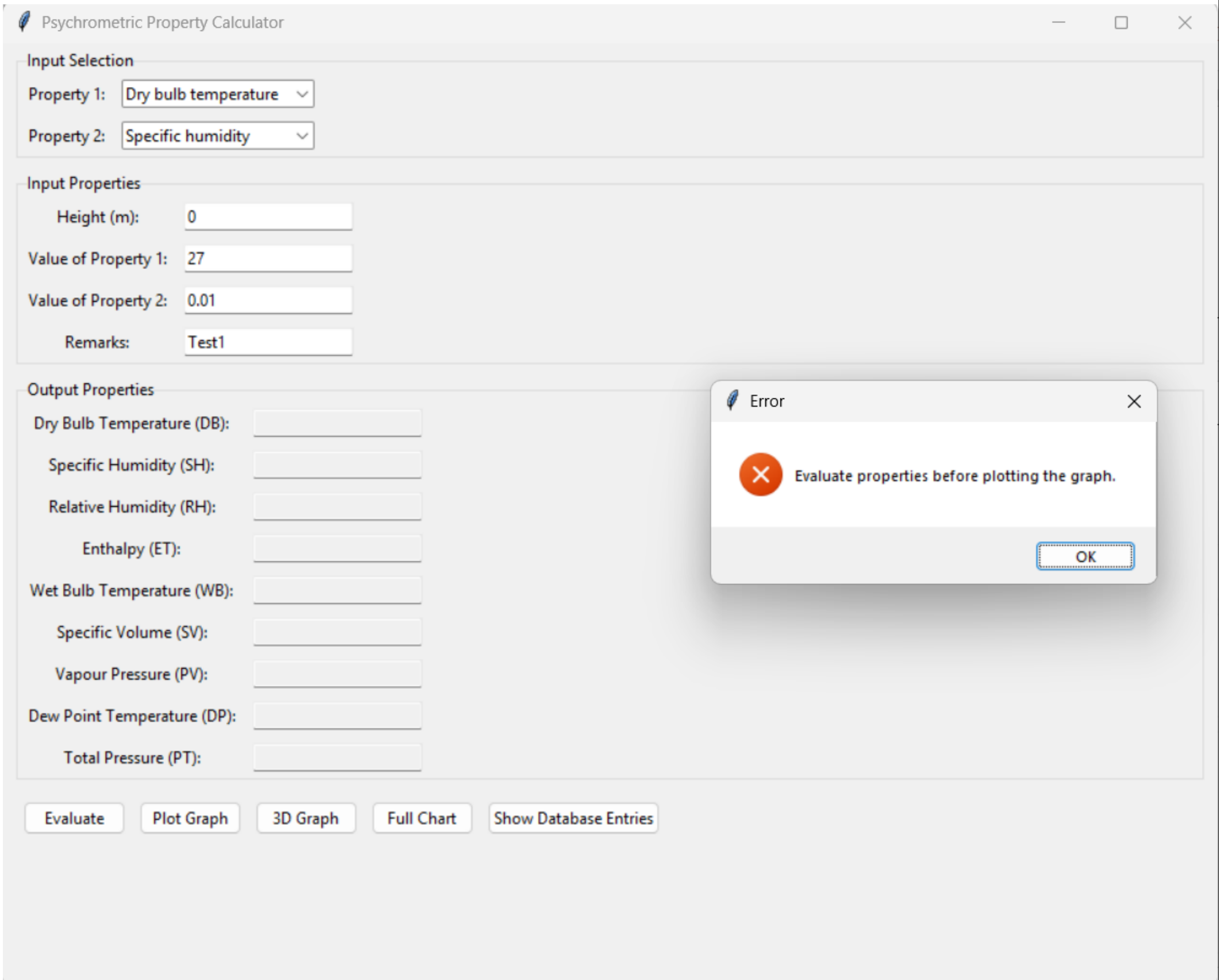
Given below are some screenshots of the working of the program:

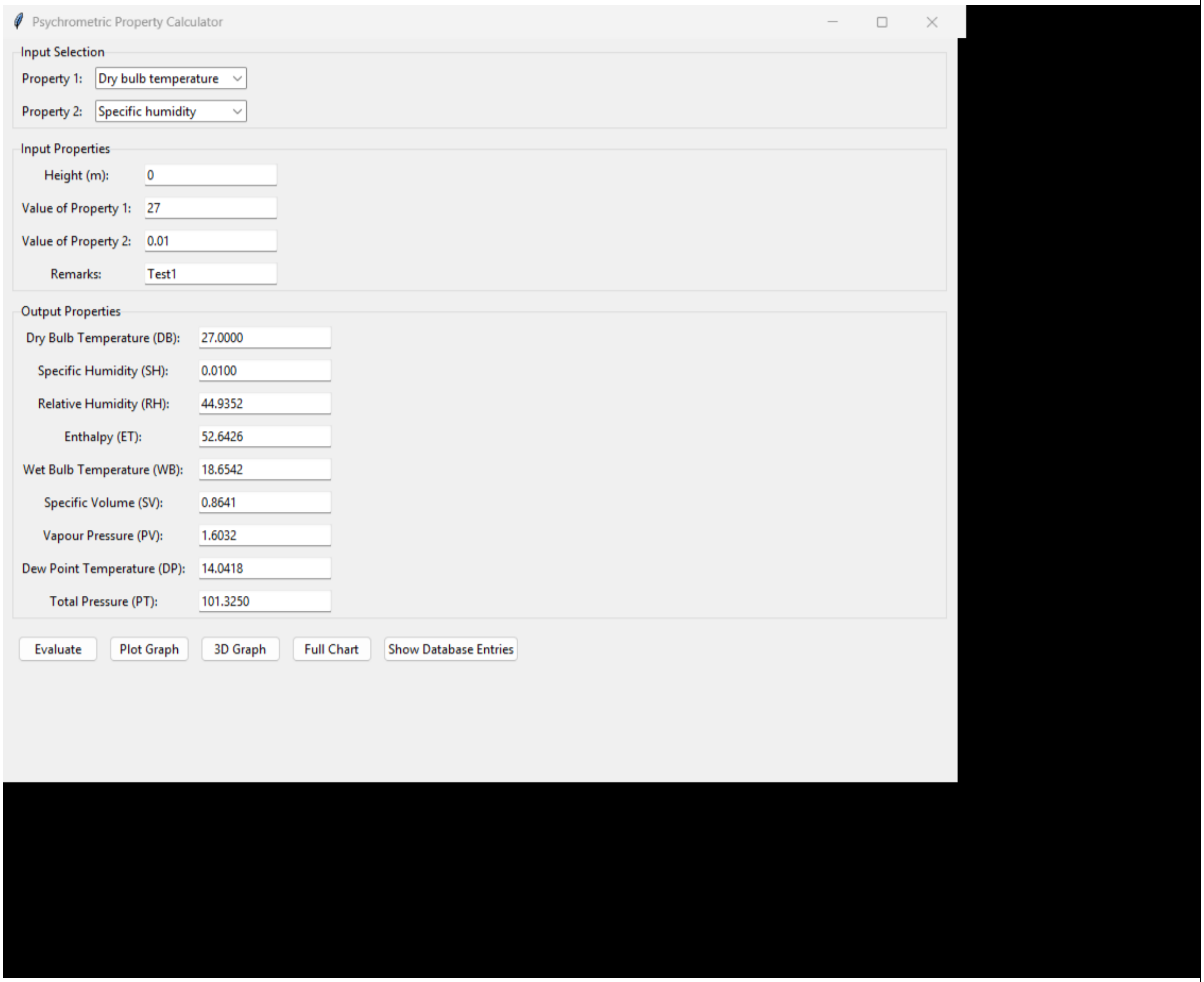

Psychrometric Property Calculator
Input Selection
Property 1:
Property 2:
Input Properties
Height (m):
Value of Property 1:
Value of Property 2:
Remarks:
Output Properties
Dry Bulb Temperature (DB):
Specific Humidity (SH):
Relative Humidity (RH):
Enthalpy (ET):
Wet Bulb Temperature (WB):
Specific Volume (SV):
Vapour Pressure (PV):
Dew Point Temperature (DP):
Total Pressure (PT):
Evaluate
Plot Graph
3D Graph
Full Chart
Show Database Entries


Psychrometric Property Calculator
Input Selection
Property 1:
Wet bulb temperature
Property 2:
Enthalpy
Input Properties
Height (m):
0
Value of Property 1:
27
Value of Property 2:
50
Remarks:
Test2
Output Properties
Dry Bulb Temperature (DB):
Specific Humidity (SH):
Relative Humidity (RH):
Enthalpy (ET):
Wet Bulb Temperature (WB):
Specific Volume (SV):
Vapour Pressure (PV):
Dew Point Temperature (DP):
Total Pressure (PT):
Logic Error
Unsupported property combination.
OK
Evaluate
Plot Graph
3D Graph
Full Chart
Show Database Entries


Psychrometric Property Calculator
Input Selection
Property 1:
Dry bulb temperature
Property 2:
Specific humidity
Input Properties
Height (m):
Value of Property 1:
Value of Property 2:
Remarks:
Output Properties
Dry Bulb Temperature (DB):
Specific Humidity (SH):
Relative Humidity (RH):
Enthalpy (ET):
Wet Bulb Temperature (WB):
Specific Volume (SV):
Vapour Pressure (PV):
Dew Point Temperature (DP):
Total Pressure (PT):
Input Error
All fields must be filled.
OK
Evaluate
Plot Graph
3D Graph
Full Chart
Show Database Entries


Psychrometric Property Calculator
Input Selection
Property 1:
Dry bulb temperature
Property 2:
Specific humidity
Input Properties
Height (m):
0
Value of Property 1:
27
Value of Property 2:
0.01
Remarks:
Test1
Output Properties
Dry Bulb Temperature (DB):
Specific Humidity (SH):
Relative Humidity (RH):
Enthalpy (ET):
Wet Bulb Temperature (WB):
Specific Volume (SV):
Vapour Pressure (PV):
Dew Point Temperature (DP):
Total Pressure (PT):
Evaluate
Plot Graph
3D Graph
Full Chart
Show Database Entries


Psychrometric Property Calculator
Input Selection
Property 1:
Dry bulb temperature
Property 2:
Specific humidity
Input Properties
Height (m):
0
Value of Property 1:
27
Value of Property 2:
0.01
Remarks:
Test1
Output Properties
Dry Bulb Temperature (DB):
Specific Humidity (SH):
Relative Humidity (RH):
Enthalpy (ET):
Wet Bulb Temperature (WB):
Specific Volume (SV):
Vapour Pressure (PV):
Dew Point Temperature (DP):
Total Pressure (PT):
Error
Evaluate properties before plotting the graph.
OK
Evaluate
Plot Graph
3D Graph
Full Chart
Show Database Entries


Psychrometric Property Calculator
Input Selection
Property 1:
Dry bulb temperature
Property 2:
Specific humidity
Input Properties
Height (m):
0
Value of Property 1:
27
Value of Property 2:
0.01
Remarks:
Test1
Output Properties
Dry Bulb Temperature (DB):
27.0000
Specific Humidity (SH):
0.0100
Relative Humidity (RH):
44.9352
Enthalpy (ET):
52.6426
Wet Bulb Temperature (WB):
18.6542
Specific Volume (SV):
0.8641
Vapour Pressure (PV):
1.6032
Dew Point Temperature (DP):
14.0418
Total Pressure (PT):
101.3250
Evaluate
Plot Graph
3D Graph
Full Chart
Show Database Entries

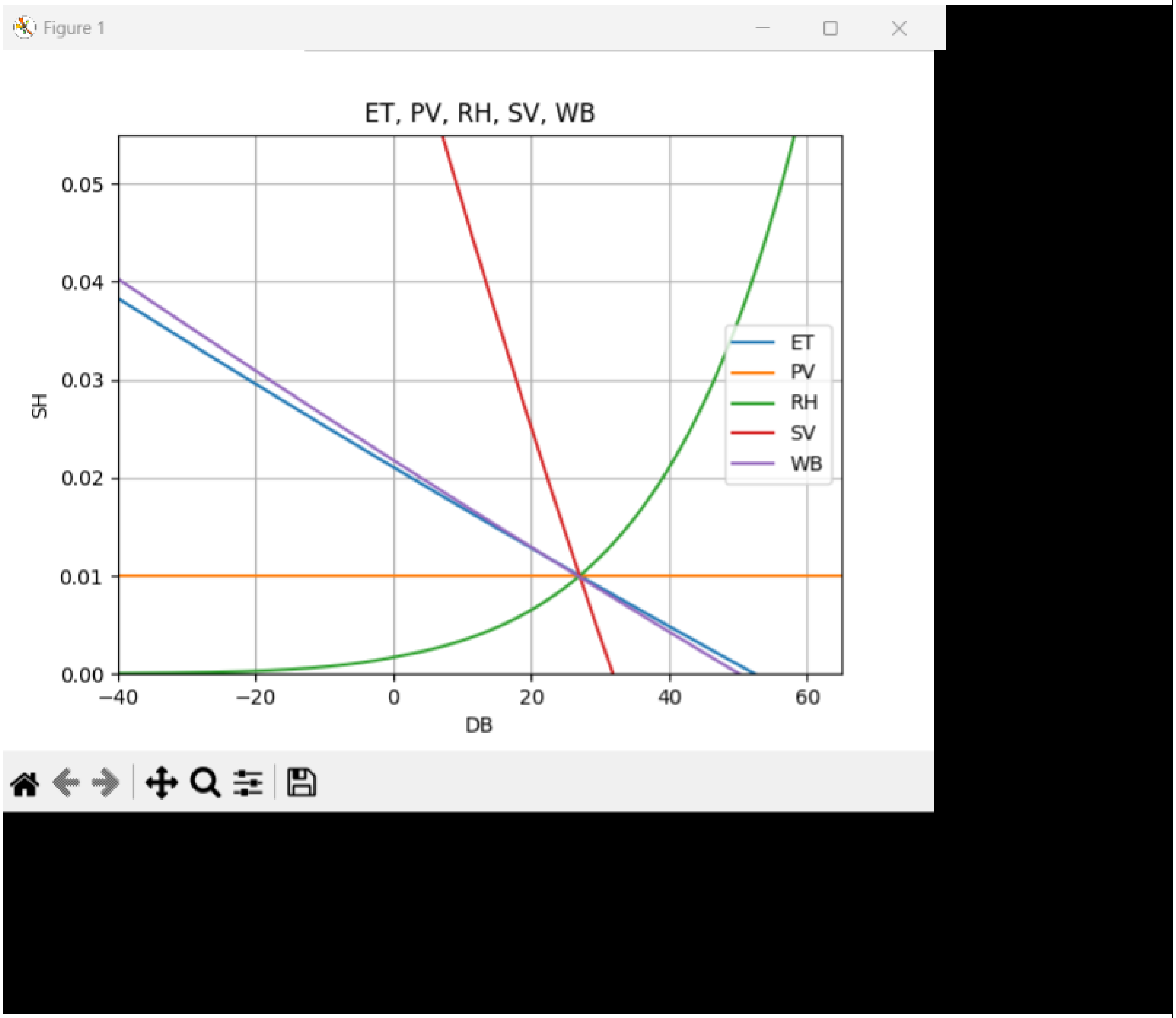
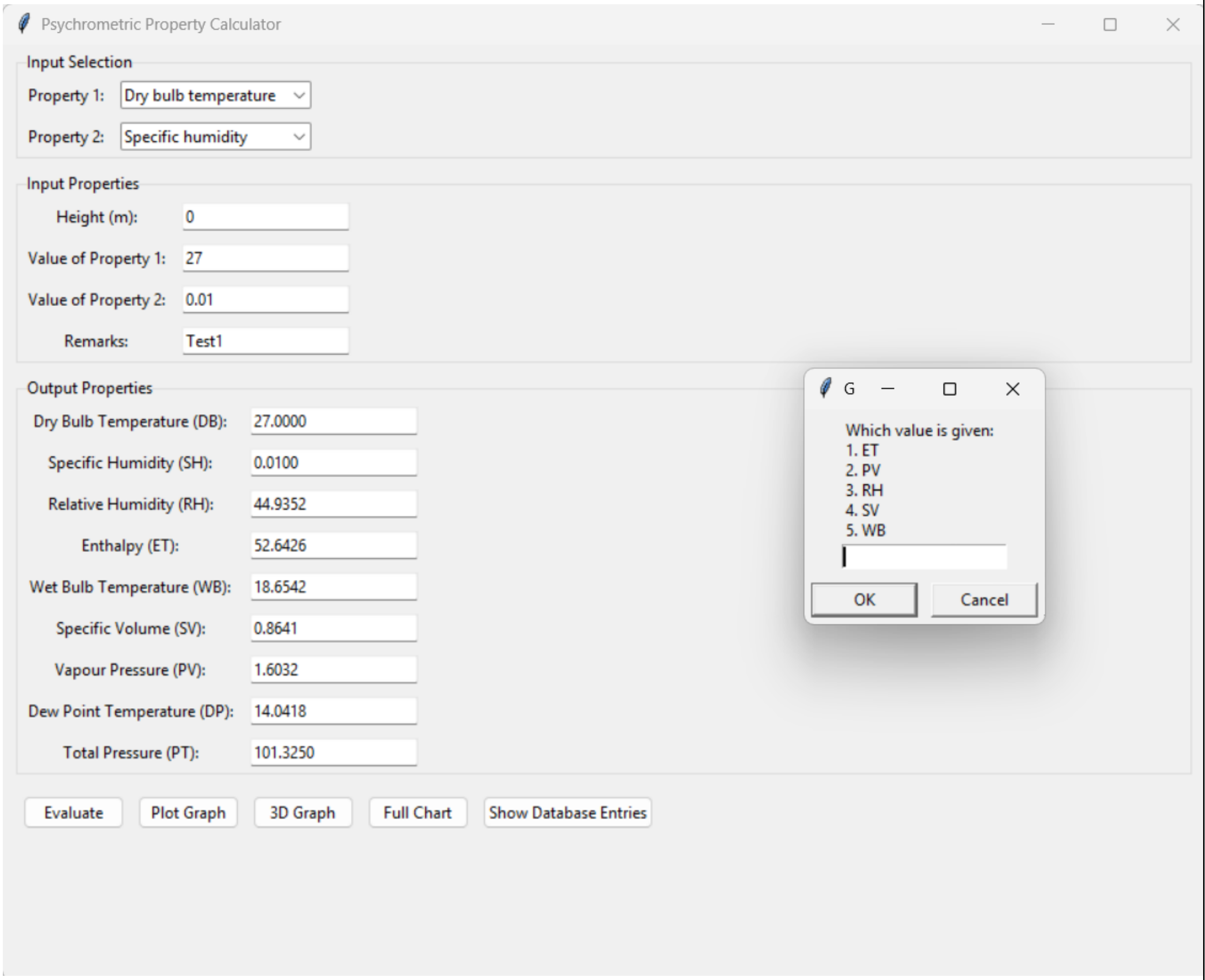
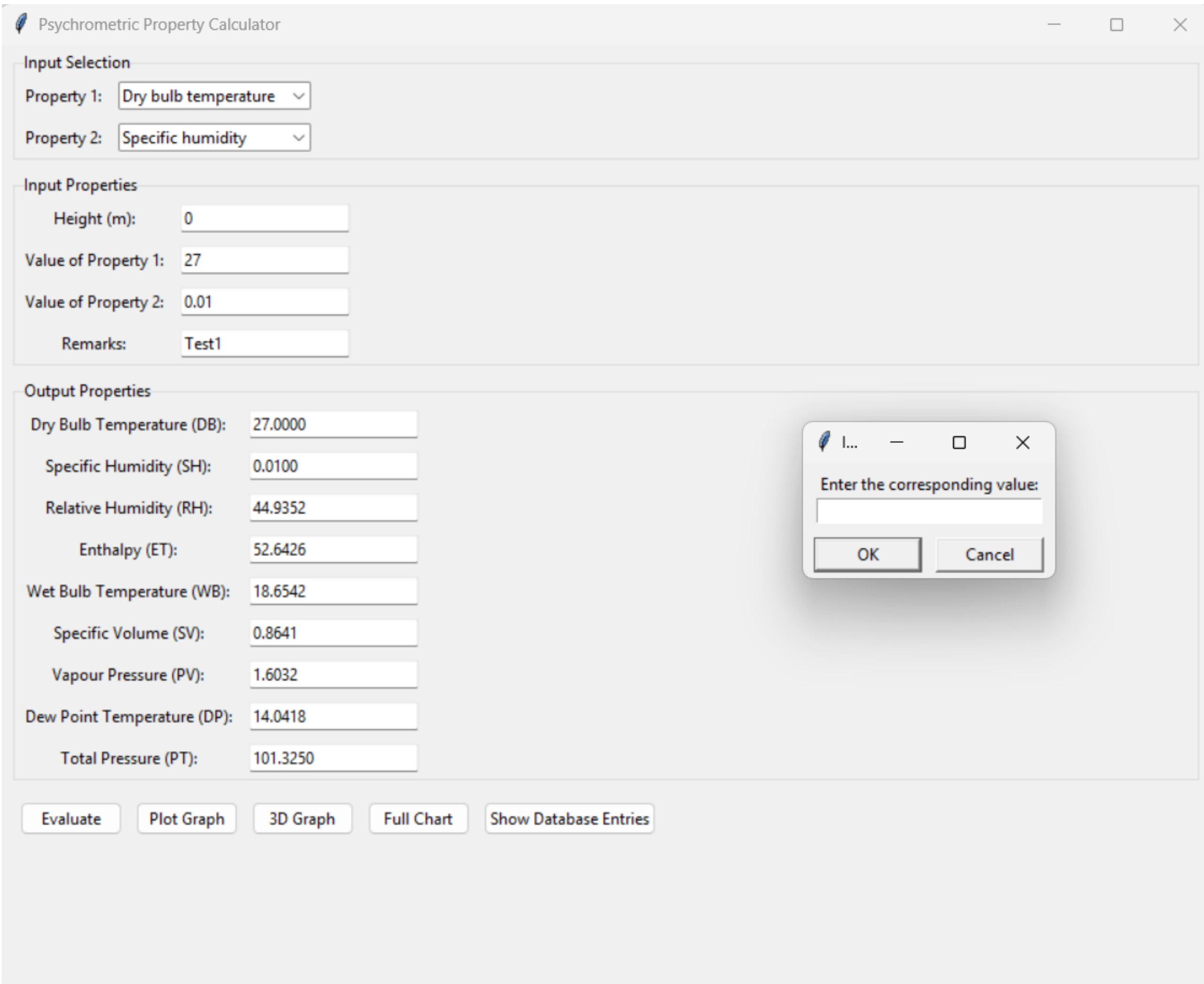


Figure 1
ET, PV, RH, SV, WB
SH
0.05
0.04
0.03
0.02
0.01
0.00
−40
−20
0
20
40
60
DB
ET
PV
RH
SV
WB


Psychrometric Property Calculator
Input Selection
Property 1: Dry bulb temperature
Property 2: Specific humidity
Input Properties
Height (m): 0
Value of Property 1: 27
Value of Property 2: 0.01
Remarks: Test1
Output Properties
Dry Bulb Temperature (DB): 27.0000
Specific Humidity (SH): 0.0100
Relative Humidity (RH): 44.9352
Enthalpy (ET): 52.6426
Wet Bulb Temperature (WB): 18.6542
Specific Volume (SV): 0.8641
Vapour Pressure (PV): 1.6032
Dew Point Temperature (DP): 14.0418
Total Pressure (PT): 101.3250
Evaluate
Plot Graph
3D Graph
Full Chart
Show Database Entries
G
Which value is given:
1. ET
2. PV
3. RH
4. SV
5. WB
OK
Cancel


Psychrometric Property Calculator
Input Selection
Property 1:
Dry bulb temperature
Property 2:
Specific humidity
Input Properties
Height (m):
0
Value of Property 1:
27
Value of Property 2:
0.01
Remarks:
Test1
Output Properties
Dry Bulb Temperature (DB):
27.0000
Specific Humidity (SH):
0.0100
Relative Humidity (RH):
44.9352
Enthalpy (ET):
52.6426
Wet Bulb Temperature (WB):
18.6542
Specific Volume (SV):
0.8641
Vapour Pressure (PV):
1.6032
Dew Point Temperature (DP):
14.0418
Total Pressure (PT):
101.3250
Evaluate
Plot Graph
3D Graph
Full Chart
Show Database Entries
I...
Enter the corresponding value:
OK
Cancel

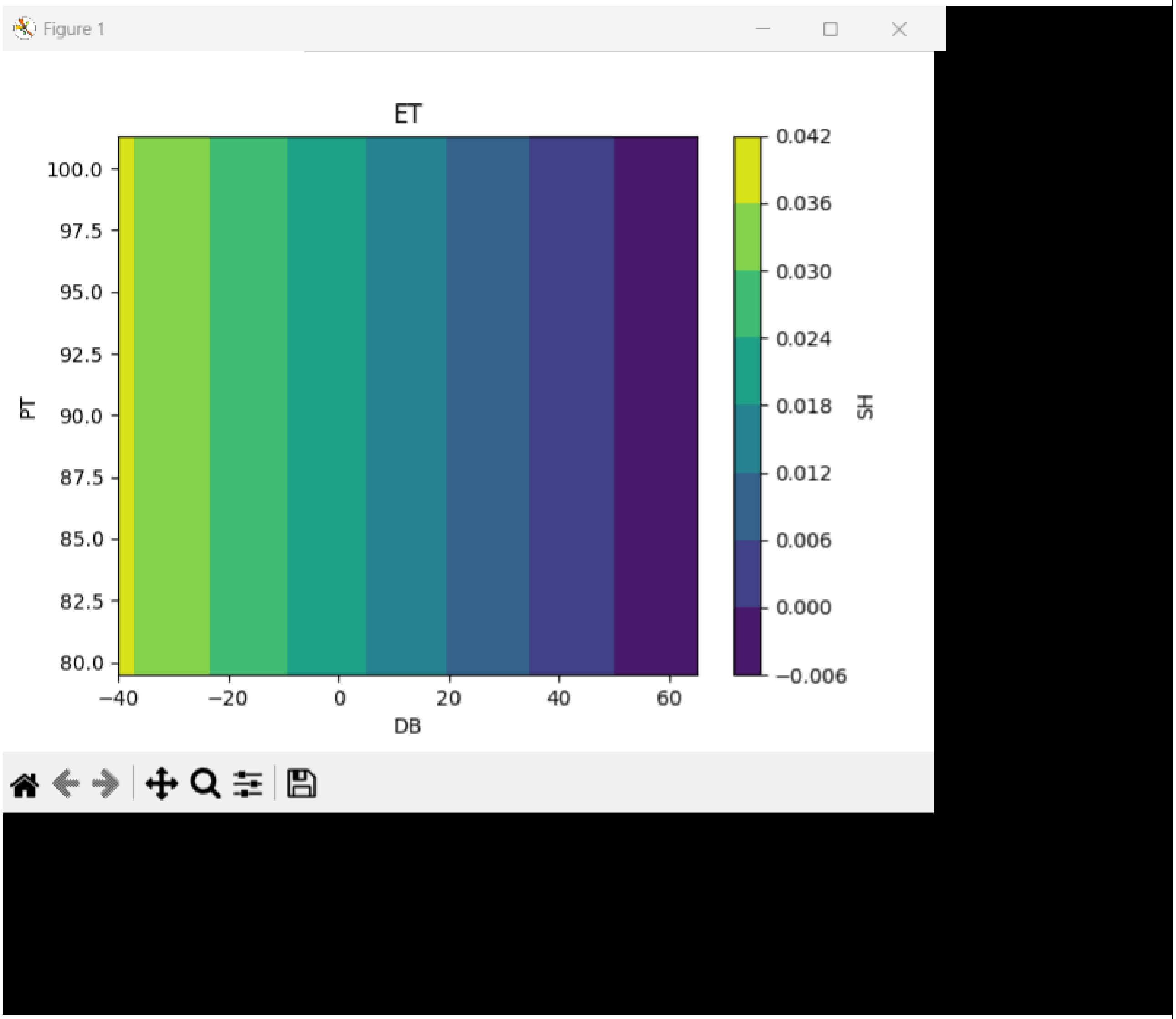
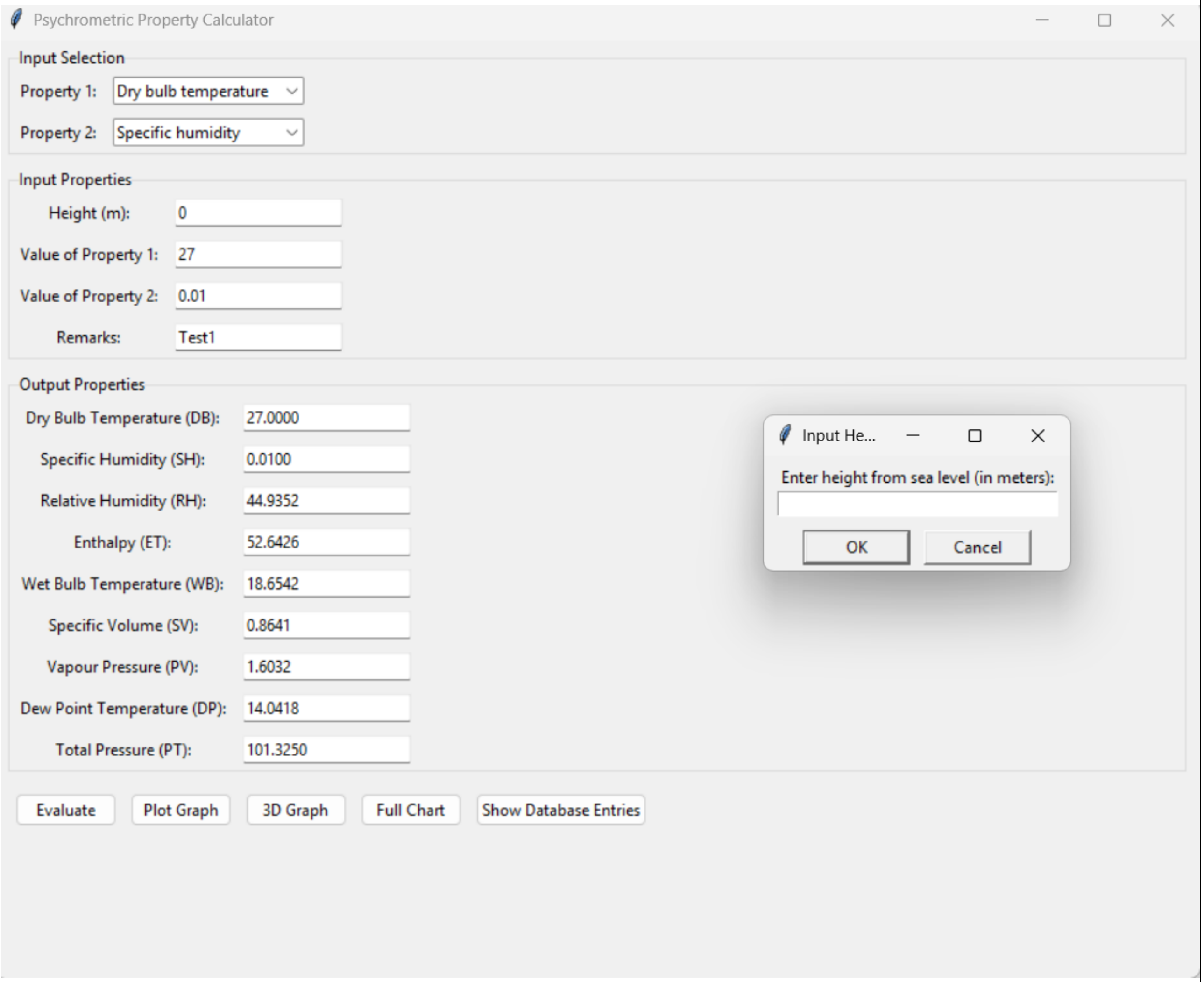
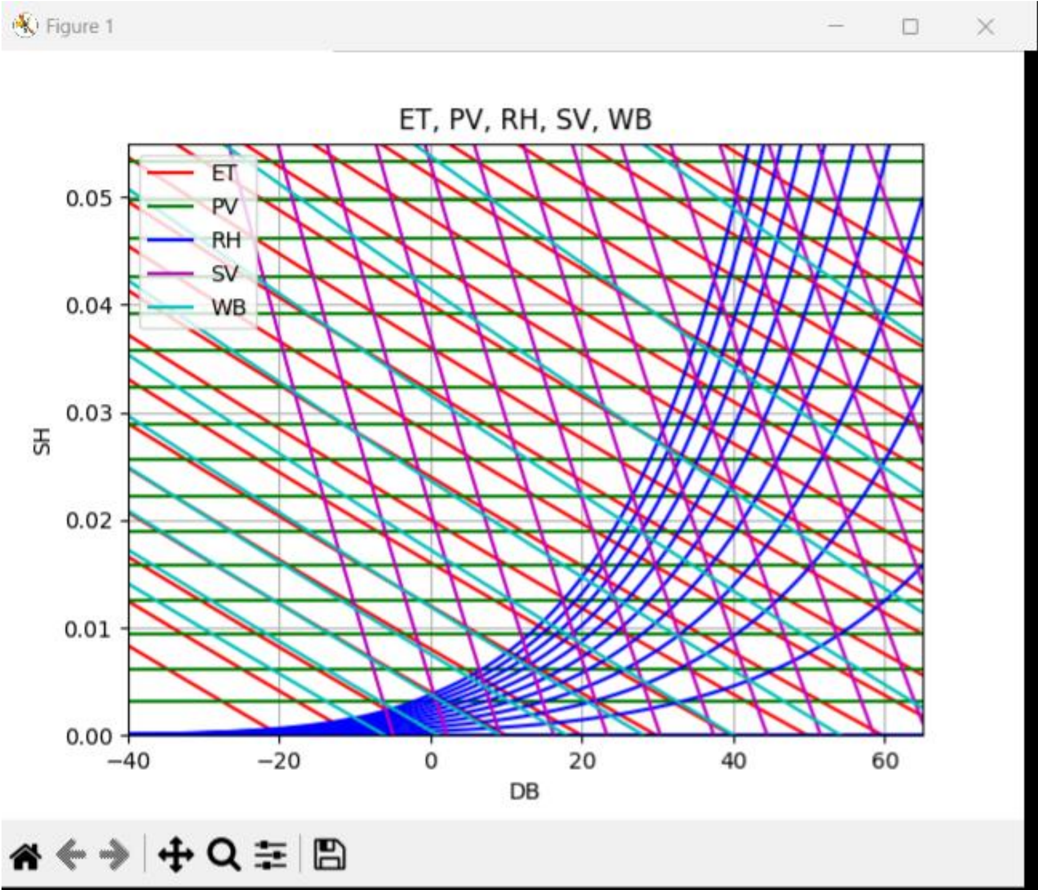


Figure 1
ET
100.0
97.5
95.0
92.5
90.0
87.5
85.0
82.5
80.0
PT
−40
−20
0
20
40
60
DB
0.042
0.036
0.030
0.024
0.018
0.012
0.006
0.000
−0.006
SH


Psychrometric Property Calculator
Input Selection
Property 1:
Dry bulb temperature
Property 2:
Specific humidity
Input Properties
Height (m):
0
Value of Property 1:
27
Value of Property 2:
0.01
Remarks:
Test1
Output Properties
Dry Bulb Temperature (DB):
27.0000
Specific Humidity (SH):
0.0100
Relative Humidity (RH):
44.9352
Enthalpy (ET):
52.6426
Wet Bulb Temperature (WB):
18.6542
Specific Volume (SV):
0.8641
Vapour Pressure (PV):
1.6032
Dew Point Temperature (DP):
14.0418
Total Pressure (PT):
101.3250
Input He...
Enter height from sea level (in meters):
OK
Cancel
Evaluate
Plot Graph
3D Graph
Full Chart
Show Database Entries




Database Entries

| DB | SH | RH | ET | WB | SV | PV | DP | PT | Remarks | TimeStamp |
|---|---|---|---|---|---|---|---|---|---|---|
| 27.0 | 0.01 | 44.9352 | 52.6426 | 18.6542 | 0.864135 | 1.60324 | 14.0418 | 101.325 | Test1 | 2024-12-17 16:56: |

# Chapter-4

## Future Plan

Implementation of GUI added more usability to the program making the program more accessible and user friendly.

But it can be improved further by making a web application out of it.

Here is my plan:

1. Make a web application of the program using Node Js as backend and MySQL as database.
2. No login is required, but while the user is at the web page, the evaluated data will be added to the temporary database and will be deleted once the user refreshes the web page.
3. Upload the project to GitHub.

# Conclusion

This project focused on developing a robust tool for calculating psychrometric properties, enhancing its usability and applicability for optimizing Atmospheric Water Generator (AWG) systems. To overcome limitations in traditional formulae, ASHRAE tables were implemented for calculating saturation temperature and pressure, eliminating errors for temperatures below 0°C. The use of PCHIP interpolation ensured precise and reliable results across a wider range of environmental conditions.

The program's iterative analysis was improved by resolving convergence issues and refining functions, making calculations more efficient. By integrating ASHRAE tables into iterative processes, the program became more concise and robust, ensuring consistency and accuracy in results.

To enhance usability, a Graphical User Interface (GUI) was developed by converting the MATLAB code into Python. The GUI simplifies property selection, validates inputs, and calculates psychrometric properties with a user-friendly interface. It also integrates data management features, automatically storing calculations with timestamps and remarks in a database. Visualization capabilities were expanded with options to generate constant property curves, 3D graphs, and full psychrometric charts, providing intuitive insights into air properties.

This enhanced tool is now a valuable asset for AWG design and optimization, with plans for further improvements, including refined 3D graphing and real-time data integration.

# References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5. Kreider, J., Curtiss, P., & Rabl, A. (2000). *Heating and Cooling of Buildings: Design for Efficiency*. CRC Press.